

A STYLE MODEL INDUCTION



STYLE REFERENCE
**Small,
high-quality
dialogue
corpus**

1

Dialogue Feature Analysis

states • strate-
gies • transitions

dialogue features

2

Bayesian Dialogue Modeling

transition & obser-
vation probabilities

B DATA SELECTION & ADAPTATION



CANDIDATE POOL
**Large, genera
purpose
dialogue corpus**

candidate responses

3

Style Com- patibility Scoring

$\text{Score}(r_t) \in [0, 1]$

Bayesian dialogue model

scored candidates

4

Preference Data Selection & DPO Adaptation

chosen $\succ$ *rejected*

LoRA-DPO



**Style-adapted
local LLM**